# MATTHEW ZHOU

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## *EDUCATION*

**Georgia Institute of Technology**, Atlanta, GA

- Candidate for Bachelor of Science in Mechanical Engineering and Minor in German (GPA: 4.00/4.00) [Aug. 2023 – May. 2027 (Expected)]

**International Department, Affiliated High School of South China Normal University**, Guangzhou, China

- Cumulative GPA: 4.94/5.00 [Aug. 2020 – Jun. 2023]
- Academic Excellence Award (¥ 2,000 Scholarship)

## *PROJECT & RESEARCH*

**Independent Gyrotourbillon Design**, Guangzhou, China

Personal Project [May. 2023 – Jun. 2023]

- Learned escapement mechanism and basics of tourbillon online. Iterated to 3D-printed final version in 1 month (CAD Models).
- Employed a bold design of concentric escapement wheel and hairspring to save space and decrease complexity.

**Automatic Solar-Powered Irrigation System Design & Construction (MAKERs' Aid Project for an Allied Club)**, Guangzhou, China

Team Project [Mar. 2023 – May. 2023]

- Developed to care for tomato plants through the summer break, providing regular irrigation to the plants without human intervention.
- Designed all 3D-printed parts and circuits, including two subsystems with Arduino Nano boards controlling moisture sensors and water pumps.
- Overcame challenges including code debugging and charging circuit malfunction.

**Material Failure Studies & Metal Fatigue Analysis**, Online

Pioneer Research Program with Prof. Michael McInerney from Rose-Hulman Institute of Technology [Jun. 2022 – Sep. 2022]

- Studied material failure types including rupture, creep, fatigue, and corrosion, with a particular focus on fatigue. Final Grade A+ (Research Paper).
- Developed a programmed machine with 3D-printed parts to experiment with metal fatigue in solder strings, which impressed the professor.

**Butane Gun Development (for MAKERs' Warship Project)**, Guangzhou, China

Personal Project/Team Course Project [Dec. 2021 – May. 2022]

- Learned fuel combustion and ballistics. Final version featured efficient 3D-printed combustion chamber and rifling that optimized performance.
- Conducted research investigating the impact of rifling pitch and barrel length on penetration (Research Paper & Presentation to Physics Club).

## *EXTRACURRICULAR ACTIVITY*

**Marine Robotics Group at Georgia Tech**, Atlanta, GA [Sep. 2023 – Present]

Mechanical Subteam Member

- Work in the Robosub Project and handle one design of the gripper subsystem.
- Kept full attendance in work and meeting sessions and joined reorganization of the lab layout.

**RoboRambler at Georgia Tech**, Atlanta, GA [Sep. 2023 – Present]

Mechanical Subteam Member

- Mended the severely impaired robot from the 2023 summer game.
- Pushed forward the chartering process of our team as a registered student organization at Georgia Tech and actively reached out to recruits.

**HFI MAKERs' & Developers' Community (MAKERs')**, Guangzhou, China [Aug. 2021 – May. 2023]

Co-Founder & Vice President

- Developed a comprehensive hardware curriculum including CAD modeling, 3D printing, electronics, etc.
- Managed administrative, financial, and public-relational affairs, implementing a streamlined management system.
- Helped developed a school utility website featuring a reservation system, lost-n-found system, and club main page (https://www.gdhfi-uc.com/).

**FIRST Robotics Competition**, Guangzhou, China [Jul. 2021 – Dec. 2022]

Co-Leader & Driver**,** Team 8214

- Directed the construction and maintenance of the team's robot, planned the most wiring layout, and designed the intake and the ball hopper.
- Judges Award received for robot design and community influence (MAKERs').

## *WORK EXPERIENCE*

**Wargaming,** Cyprus [Jul. 2022 – Present]

Supertester for *World of Warships*, Asian Realm

- Collaborate with a global team of Supertesters to test unreleased content during monthly testing cycles
- Generate detailed bug reports and provide feedback to developers with valuable insights.

## *SKILLS*

**3D Model Software:** SolidWorks (2.5 years), Onshape (1.5 years).

**Programming:** Java, Arduino.

**Language:** Mandarin (native), English (bilingual), Cantonese (native), German (A1 Certificated)